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Signal Detection for HF Skywave Massive MIMO-OFDM with Slepian Transform

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***Abstract*—In this paper, we investigate signal detection for high frequency (HF) skywave massive multiple-input multiple-output (MIMO) systems with orthogonal frequency division multiplexing (OFDM) modulation. We first reveal the relationship of sparse supports between the beam domain channel and the Fourier spectrum of the space domain channel for HF skywave massive MIMO-OFDM channels. We then propose a separate Slepian transform (SST) based detector, where a set of modulated Slepian sequences are designed separately for user terminals (UTs). Before the MMSE detection, the Slepian transform is performed to reduce the dimension for each UT, thus avoiding high dimensional matrices multiplications and inversions. To further reduce the computational complexity, we propose a joint Slepian transform (JST) based detector, where a fixed set of modulated Slepian sequences are designed, and Slepian transforms of the observation vector can be efficiently implemented based on low-dimensional fast Fourier transform (FFT). Simulation results demonstrate the attractive performance and excellent complexity advantage of proposed detectors.**

***Index Terms*—Massive MIMO-OFDM, HF skywave communications, Slepian transform, detector.**

I. Introduction

The high frequency (HF) band communication, whose frequency range is typically from 3 to 30 MHz, can be utilized for over-the-horizon communications by ionosphere refraction. For global coverage, satellite communication is a research hotspot. In contrast, HF skywave communication has a comparatively low deployment cost but suffers from low data rates because of the scarce spectrum resource and challenging ionosphere conditions [1]. By deploying a large number of antennas at the base station (BS) to serve enormous user terminals (UTs) simultaneously [2], massive multiple-input multiple-output (MIMO) has been exploited to significantly enhance the system sumrate and spectrum efficiencies of wireless cellular communications and has been introduced into HF skywave communications more recently [3].

However, due to the increased number of BS antennas and UTs, there are several challenges for massive MIMO systems, one of which is uplink (UL) signal detection. When the dimension of the massive MIMO system is large, some widely used detectors such as sphere decoder [4], zero-forcing (ZF) detector and minimum mean-squared error (MMSE) detector suffer from high computational complexity. Motivated by the sparsity of the beam domain channel, several beam based detectors with low complexity have been proposed recently [5]–[7]. Thanks to the small angle spread in HF skywave massive MIMO scenario, the sparsity of the beam domain channel can also be exploited to design detectors.

According to the beam based HF skywave massive MIMO channel model [3], each UT occupies a few non-zero beams, implying that the space domain channel is approximately band-limited and can be compressed into a low-dimensional form. Inspired by the property of maximizing spectral concentration, Slepian sequences [8] can be utilized for dimensionality reduction.

In this paper, we investigate Slepian transform based detectors for HF skywave massive MIMO systems, where the transform matrix is constructed by modulated Slepian sequences. We first present a sparse beam based channel model and establish the relationship of sparse supports between the beam domain channel and the Fourier spectrum of the space domain channel. Then, we propose a separate Slepian transform (SST) based detector, where the Slepian transform is performed for each UT individually before the low-dimensional MMSE filtering. Further, we develop a joint Slepian transform (JST) based detector where the Slepian transform is performed jointly for UTs in a given UT set, and the Slepian transform for the observation vector can be efficiently implemented by low-dimensional fast Fourier transform (FFT). As a result, the proposed detectors can achieve comparable detection performance to the MMSE detector with much lower complexity.

II. Channel Model

A. System Configuration

We consider a HF skywave massive MIMO-OFDM system operating in time-division duplex (TDD) mode. The BS is equipped with a uniform linear array (ULA) with M antennas, and U single-antenna UTs are served. Let N_c, N_g, N_v and Δf denote the number of subcarriers, the length of the cyclic prefix (CP), the number of valid subcarriers and subcarrier spacing, respectively. Thus, the OFDM sample interval is $T_{\mathrm{sym}} = \frac{N_c+N_g}{N_c\Delta f}$. The antennas index set and UTs index set are

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represented as $\mathcal{A} = \{1, 2, \cdots, M\}$ and $\mathcal{U} = \{1, 2, \cdots, U\}$, respectively. In HF communication system, due to the complicated conditions of the ionosphere, the carrier frequency f_c varies with different ionosphere conditions, and the highest system operating frequency is defined as f_o. The inter-antenna spacing is set as $d = \lambda_o/2$, where $\lambda_o = c/f_o$ is the wavelength and c is the speed of light.

B. Beam Based Channel Model

In the uplink (UL) transmission, let $x_{u,n,k}$ denote the data transmitted by the u-th UT at the k-th subcarrier on the n-th OFDM symbol. We assume that the channel states remain constant in one OFDM symbol. After OFDM demodulation, the received signal at the k-th subcarrier on the n-th OFDM symbol at the m-th antenna is

$$y_{n,k}[m] = \sum_{u \in \mathcal{U}} h_{u,n,k}[m] x_{u,n,k} + z_{n,k}[m], \tag{1}$$

where $z_{n,k}[m] \sim \mathcal{CN}(0, \sigma_z^2)$ is the additive white Gaussian noise, $h_{u,n,k}[m]$ is the channel frequency response at the k-th subcarrier on the n-th OFDM symbol between the u-th UT and the m-th antenna at the BS, which can be given as [3]

$$h_{u,n,k}[m] = \sum_{p=1}^{P_u} \eta_{u,p,n,k} e^{-\bar{\jmath} 2\pi (f_c + k\Delta f)\Delta\tau (m-1)\Omega_{u,p}}, \tag{2}$$

where P_u is the number of paths from the u-th UT to the BS and $\Delta\tau = d/c$. Additionally,

$$\eta_{u,p,n,k} \triangleq \beta_{u,p} e^{\bar{\jmath}\phi_{u,p}} e^{\bar{\jmath} 2\pi f_{u,p}^{\mathrm{d}} n T_{\mathrm{sym}}} e^{-\bar{\jmath} 2\pi k \Delta f \tau_{u,p}}, \tag{3}$$

where $\beta_{u,p}$, $\phi_{u,p}$, $f_{u,p}^{\mathrm{d}}$, $\tau_{u,p}$ and $\Omega_{u,p}$ represent the gain, the initial phase, the Doppler frequency, the propagation delay to the first antenna at the BS and the direction cosine of the p-th path, respectively.

Next, we perform uniform sampling for direction cosine as $[-1, 1) = \bigcup_{\tilde{m}=1}^{\tilde{M}} \mathcal{S}_{\tilde{m}}$, where $\tilde{M} \geq M$ is the sampling number, $\mathcal{S}_{\tilde{m}} = [\Omega_{\tilde{m}}, \Omega_{\tilde{m}} + 2/\tilde{M})$ and $\Omega_{\tilde{m}} = (2\tilde{m} - 2 - \tilde{M})/\tilde{M}$. Define the set $\mathcal{P}_u \triangleq \{\Omega_{u,1}, \cdots, \Omega_{u,P_u}\}$, including direction cosines of all paths for UT u. Then, the space domain channel vector $\mathbf{h}_{u,n,k} \triangleq [h_{u,n,k}[1], \cdots, h_{u,n,k}[M]]^{\mathrm{T}}$ can be approximated as

$$\mathbf{h}_{u,n,k} = \sum_{\tilde{m}=1}^{\tilde{M}} \sum_{\Omega_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_{\tilde{m}}} \eta_{u,p,n,k} \mathbf{v}_k(\Omega_{\tilde{m}}) = \mathbf{V}_k \tilde{\mathbf{h}}_{u,n,k}, \tag{4}$$

where $\mathbf{V}_k = [\mathbf{v}_k(\Omega_1), \cdots, \mathbf{v}_k(\Omega_{\tilde{M}})] \in \mathbb{C}^{M \times \tilde{M}}$ is the beam matrix composed of the sampled steering vectors at the k-th subcarrier with $[\mathbf{v}_k(\Omega_{\tilde{m}})]_m = e^{-\bar{\jmath} 2\pi (f_c + k\Delta f)\Delta\tau (m-1)\Omega_{\tilde{m}}}$, and $\tilde{\mathbf{h}}_{u,n,k} \in \mathbb{C}^{\tilde{M}}$ denotes the beam domain channel vector with $[\tilde{\mathbf{h}}_{u,n,k}]_{\tilde{m}} = \sum_{\Omega_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_{\tilde{m}}} \eta_{u,p,n,k}$.

We now focus on the Fourier spectrum of $h_{u,n,k}[m]$ to design detectors with low complexity. The beam domain channel vector is sparse and associated with the space domain channel's Fourier spectrum due to the narrow angle spread from each UT to the BS [9]. We rewrite (2) as

$$h_{u,n,k}[m] = \int_{-1}^{1} \sum_{p=1}^{P_u} \eta_{u,p,n,k} e^{-\bar{\jmath} 2\pi g_k (m-1)\Omega} \delta(\Omega - \Omega_{u,p}) \mathrm{d}\Omega, \tag{5}$$

where $\delta(\cdot)$ represents the Dirac delta function, and $g_k \triangleq (f_c + k\Delta f)\Delta\tau$. As f_o is the highest system operating frequency, we have $f_c + k\Delta f < f_o$ and thus $g_k < 1/2$. Let $\nu = -g_k\Omega$, $\nu_{u,p,k} = -g_k\Omega_{u,p}$ and define

$$\tilde{h}_{u,n,k}(\nu) \triangleq \begin{cases} \sum_{p=1}^{P_u} \eta_{u,p,n,k} \delta(\nu - \nu_{u,p,k}), & \nu \in [-g_k, g_k] \\ 0, & \nu \in [-\frac{1}{2}, -g_k) \cup (g_k, \frac{1}{2}). \end{cases} \tag{6}$$

Then, (5) can be rewritten as

$$\begin{aligned} h_{u,n,k}[m] &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \tilde{h}_{u,n,k}(\nu) e^{\bar{\jmath} 2\pi (m-1)\nu} \mathrm{d}\nu \\ &= \mathcal{F}^{-1} \left\{ \tilde{h}_{u,n,k}(\nu) e^{-\bar{\jmath} 2\pi \nu} \right\}, \end{aligned} \tag{7}$$

where $\mathcal{F}^{-1}\{\cdot\}$ denotes the inverse of discrete-time Fourier transform (IDTFT). Due to the small angle spread, $h_{u,n,k}[m]$ can be modelled as a narrowband sequence. The frequency band-limit of $h_{u,n,k}[m]$ is specified as a continuous interval $\mathcal{L}_{u,k} = [\nu_{u,k}^{\mathrm{l}}, \nu_{u,k}^{\mathrm{r}}] \subset [-1/2, 1/2)$. When M tends to be large, $\nu_{u,k}^{\mathrm{l}}$ and $\nu_{u,k}^{\mathrm{r}}$ can be approximated by

$$\nu_{u,k}^{\mathrm{l}} = -g_k \left(\frac{2}{\tilde{M}} \max\{\mathcal{B}_u\} - 1 \right), \tag{8}$$

$$\nu_{u,k}^{\mathrm{r}} = -g_k \left(\frac{2}{\tilde{M}} \min\{\mathcal{B}_u\} - 1 \right), \tag{9}$$

where $\mathcal{B}_u = \{\tilde{m} | [\tilde{\mathbf{h}}_{u,n,k}]_{\tilde{m}} \neq 0\} = \{\tilde{m} | [\tilde{\boldsymbol{\omega}}_u]_{\tilde{m}} \neq 0\}$ is the index set for non-zero elements of the beam domain channel vector with $B_u = |\mathcal{B}_u|$, and $\tilde{\boldsymbol{\omega}}_u \triangleq \mathbb{E}\{\tilde{\mathbf{h}}_{u,n,k} \odot \tilde{\mathbf{h}}_{u,n,k}^*\}$ is referred to as the channel coupling vector with $[\tilde{\boldsymbol{\omega}}_u]_{\tilde{m}} = \sum_{\Omega_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_{\tilde{m}}} \beta_{u,p}^2$, which is independent of symbols and subcarriers, and is also regarded as the statistical channel state information (CSI), where $\mathbb{E}\{\cdot\}$ and $\odot$ denote the expectation and the Hadamard operator, respectively.

III. Separate Slepian Transform Based Detector

A. MMSE Detector

We assume that the perfect CSI is known at the BS and focus on the signal detection at the k-th subcarrier on the n-th OFDM symbol. For brevity, indexes n and k in the subscript are omitted hereafter. Then (1) can be rewritten in vector form

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{z}, \tag{10}$$

where $\mathbf{y} \triangleq [y[1], \cdots, y[M]]^{\mathrm{T}} \in \mathbb{C}^M$, $\mathbf{H} \triangleq [\mathbf{h}_1, \cdots, \mathbf{h}_U] \in \mathbb{C}^{M \times U}$, $\mathbf{x} \triangleq [x_1, \cdots, x_U]^{\mathrm{T}} \in \mathbb{C}^U$ and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}_M, \sigma_{\mathrm{z}}^2 \mathbf{I}_M)$ are the observation vector, the channel matrix, the transmit vector and the complex Gaussian noise vector, respectively. $\mathbf{I}_M$ represents M-dimensional identity matrix and $\mathbb{E}\{\mathbf{x}\mathbf{x}^{\mathrm{H}}\} = \sigma_{\mathrm{x}}^2 \mathbf{I}_U$, where σ_{x}^2 is the average transmit power of each UT.

The MMSE detector is given as

$$\mathbf{R}^{\mathrm{mmse}} = \mathbf{H} \left(\mathbf{H}^{\mathrm{H}} \mathbf{H} + \frac{1}{\rho} \mathbf{I}_U \right)^{-1}, \tag{11}$$

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where $\rho = \sigma_{\mathrm{x}}^2/\sigma_{\mathrm{z}}^2$, and the detection result is $\hat{\mathbf{x}}^{\mathrm{mmse}} = (\mathbf{R}^{\mathrm{mmse}})^{\mathrm{H}}\mathbf{y}$.

When M and U are both large in the HF skywave massive MIMO scenario, calculations of $\mathbf{R}^{\mathrm{mmse}}$ and $\hat{\mathbf{x}}^{\mathrm{mmse}}$ are computationally demanding due to high-dimensional matrix inversion and multiplication. Note that the channel energy for each UT is concentrated on non-zero beams, which implies that the sparsity of the beam domain channel can be exploited to compress the channel matrix and the observation vector into low-dimensional ones. By dimensionality reduction, the computational complexity of the MMSE detection can be reduced. However, the spectral leakage caused by the limited antenna number at the BS should be taken into consideration. Interestingly, Slepian sequences maximize spectral concentration among fixed length sequences, which can effectively suppress the impact of spectral leakage compared to the discrete Fourier transform (DFT) basis. This motivates us to utilize Slepian sequences for detector design. Before proceeding, Slepian sequences are introduced in the sequel.

B. Slepian Sequences

Define a discrete sequence $s[n]$ with all zeros outside the index range $0, 1, \cdots, M-1$. We expect that most of its energy is concentrated in Fourier spectrum interval $[-W, +W]$ with $W < 1/2$, i.e., the following is maximized [8]

$$\lambda = \frac{\int_{-W}^{+W} |\mathcal{F}\{s[n]\}|^2 \mathrm{d}\nu}{\int_{-\frac{1}{2}}^{\frac{1}{2}} |\mathcal{F}\{s[n]\}|^2 \mathrm{d}\nu} = \frac{\mathbf{s}^{\mathrm{H}}\mathbf{B}\mathbf{s}}{\mathbf{s}^{\mathrm{H}}\mathbf{s}}, \tag{12}$$

where $\mathbf{s} \triangleq [s[0], \cdots, s[M-1]]^{\mathrm{T}}$, and the Hermitian matrix $\mathbf{B} \in \mathbb{R}^{M\times M}$ with $[\mathbf{B}]_{i,j} \triangleq 2W\mathrm{sinc}(2W(i-j))$ is known as prolate matrix, where $\mathrm{sinc}(2W(i-j)) \triangleq \sin(2\pi W(i-j))/2\pi W(i-j)$ is the sinc function. Interestingly, (12) is the Rayleigh quotient, whose maximum value is the maximum eigenvalue of $\mathbf{B}$. To this end, we decompose $\mathbf{B}$ into $\mathbf{B} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{\mathrm{H}}$, where $\mathbf{S} \triangleq [\mathbf{s}_1, \cdots, \mathbf{s}_M] \in \mathbb{R}^{M\times M}$ consists of eigenvectors $\mathbf{s}_1, \cdots, \mathbf{s}_M$ (defined as Slepian sequences), and $\mathbf{\Lambda} \triangleq \mathrm{diag}\{[\lambda_1, \cdots, \lambda_M]\} \in \mathbb{R}^{M\times M}$ in which $\lambda_1, \cdots, \lambda_M \in (0, 1)$ are eigenvalues in descending order and $\mathrm{diag}\{\mathbf{a}\}$ indicates the diagonal matrix with $\mathbf{a}$ along its main diagonal.

Moreover, to describe a sequence with the majority of its energy contained in the Fourier spectrum interval $\mathcal{M}(\overline{\nu}) = [\overline{\nu} - W, \overline{\nu} + W]$ with $\overline{\nu} \neq 0$, modulated Slepian sequences are proposed. The modulated vector and the matrix composed of modulated Slepian sequences are respectively given as [10]

$$\boldsymbol{\omega}(\overline{\nu}) \triangleq \left[1, e^{\bar{\jmath}2\pi\overline{\nu}}, \cdots, e^{\bar{\jmath}2\pi\overline{\nu}(M-1)}\right]^{\mathrm{T}} \in \mathbb{C}^M, \tag{13}$$

$$\mathbf{D}(\overline{\nu}) \triangleq [\mathbf{d}_1(\overline{\nu}), \cdots, \mathbf{d}_K(\overline{\nu})] \in \mathbb{C}^{M\times K}, \tag{14}$$

where $\mathbf{d}_i(\overline{\nu}) \triangleq \boldsymbol{\omega}(\overline{\nu}) \odot \mathbf{s}_i \in \mathbb{C}^M$. It is well known that when $K \approx 2MW$, $\mathbf{D}(\overline{\nu})$ is sufficient to capture the sequence $\mathbf{a}$ of length M whose energy is mostly concentrated in $\mathcal{M}(\overline{\nu})$, i.e., $\mathbf{D}(\overline{\nu})\mathbf{D}(\overline{\nu})^{\mathrm{H}}\mathbf{a} \approx \mathbf{a}$. Therefore, we can compress $\mathbf{a}$ by obtaining Slepian basis coefficients $\overline{\mathbf{a}}(\overline{\nu}) = \mathbf{D}(\overline{\nu})^{\mathrm{H}}\mathbf{a}$, and we denote $\overline{\mathbf{a}}(\overline{\nu})$ as the Slepian transform of $\mathbf{a}$, which is utilized in the subsequent detector designs.

C. Separate Slepian Transform Based Detector

In this subsection, the Slepian transform is exploited for detection. We construct $\mathbf{D}(\overline{\nu}_u)$ with $\mathcal{M}(\overline{\nu}_u) = [\overline{\nu}_u - W, \overline{\nu}_u + W]$ for each UT u (the parameter settings of $\mathbf{D}(\overline{\nu}_u)$ are discussed later), then perform the Slepian transform to $\mathbf{y}$ as

$$\overline{\mathbf{y}}(\overline{\nu}_u) = \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{y} = \sum_{u'\in\mathcal{U}} \overline{\mathbf{h}}_{u'}(\overline{\nu}_u)x_{u'} + \overline{\mathbf{z}}(\overline{\nu}_u), \tag{15}$$

where $\overline{\mathbf{y}}(\overline{\nu}_u) \in \mathbb{C}^K$ indicates the compressed observation vector, $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u) \triangleq \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{h}_{u'} \in \mathbb{C}^K$ represents the compressed channel vector and $\overline{\mathbf{z}}(\overline{\nu}_u) \triangleq \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{z} \in \mathbb{C}^K$ denotes the compressed noise vector. Note that $\mathbf{D}(\overline{\nu}_u)$ is composed of the first K columns of a unitary matrix, thus $\overline{\mathbf{z}}(\overline{\nu}_u) \sim \mathcal{CN}(0, \sigma_z^2\mathbf{I}_K)$.

With the compressed observation vector in (15), the MMSE detector of UT u is given by

$$\begin{aligned}\mathbf{r}_u^{\mathrm{sst}} &= \arg\min_{\mathbf{r}_u\in\mathbb{C}^K} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\mathrm{H}}\overline{\mathbf{y}}(\overline{\nu}_u)\right|^2\right\} \\ &= \left(\sum_{u'\in\mathcal{U}} \overline{\mathbf{h}}_{u'}(\overline{\nu}_u)\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)^{\mathrm{H}} + \frac{1}{\rho}\mathbf{I}_K\right)^{-1} \overline{\mathbf{h}}_u(\overline{\nu}_u). \end{aligned}\tag{16}$$

Consequently, the detection result is

$$\hat{\mathbf{x}}^{\mathrm{sst}} = \left[\left(\mathbf{r}_1^{\mathrm{sst}}\right)^{\mathrm{H}}\overline{\mathbf{y}}(\overline{\nu}_1), \cdots, \left(\mathbf{r}_U^{\mathrm{sst}}\right)^{\mathrm{H}}\overline{\mathbf{y}}(\overline{\nu}_U)\right]^{\mathrm{T}} = \left(\mathbf{R}^{\mathrm{sst}}\right)^{\mathrm{H}}\mathbf{y}, \tag{17}$$

where

$$\mathbf{R}^{\mathrm{sst}} = \left[\mathbf{D}(\overline{\nu}_1)\mathbf{r}_1^{\mathrm{sst}}, \cdots, \mathbf{D}(\overline{\nu}_U)\mathbf{r}_U^{\mathrm{sst}}\right] \in \mathbb{C}^{M\times U}. \tag{18}$$

Generally, the detection performance of (17) is inferior to that of the MMSE detector due to the fact

$$\begin{aligned}\mathbb{E}\left\{\|\mathbf{x} - \hat{\mathbf{x}}^{\mathrm{sst}}\|^2\right\} &= \sum_{u\in\mathcal{U}} \min_{\mathbf{r}_u\in\mathbb{C}^K} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\mathrm{H}}\overline{\mathbf{y}}(\overline{\nu}_u)\right|^2\right\} \\ &= \sum_{u\in\mathcal{U}} \min_{\mathbf{r}_u\in\mathrm{span}\{\mathbf{D}(\overline{\nu}_u)\}} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\mathrm{H}}\mathbf{y}\right|^2\right\} \\ &\geq \sum_{u\in\mathcal{U}} \min_{\mathbf{r}_u\in\mathbb{C}^M} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\mathrm{H}}\mathbf{y}\right|^2\right\} \\ &= \mathbb{E}\left\{\|\mathbf{x} - \hat{\mathbf{x}}^{\mathrm{mmse}}\|^2\right\}, \end{aligned}\tag{19}$$

where $\mathrm{span}\{\mathbf{D}(\overline{\nu}_u)\}$ represents the column space of $\mathbf{D}(\overline{\nu}_u)$. And the computation of $\mathbf{r}_u^{\mathrm{sst}}$ is still sophisticated because all UTs are considered as inter-UT interference in (16). Accordingly, to guarantee the detection performance and simplify (16), we have to design appropriate parameters for $\mathbf{D}(\overline{\nu}_u)$. To address this issue, we develop the following theorems, which provide the criterion for designing K, W and $\overline{\nu}_u$. The proofs are omitted due to space limitations.

Theorem 1: Assume that $\mathcal{P}_u \cap \mathcal{P}_{u'} = \emptyset$ for $u \neq u'$. When $\mathcal{L}_u \subseteq \mathcal{M}(\overline{\nu}_u)$, $K = \lfloor 2MW(1+\epsilon)\rfloor$, $0 < \epsilon < \frac{1}{2W} - 1$, we have

$$\lim_{M\to\infty} \mathbb{E}\left\{\|\hat{\mathbf{x}}^{\mathrm{mmse}} - \hat{\mathbf{x}}^{\mathrm{sst}}\|^2\right\} = 0, \tag{20}$$

where $\lfloor x\rfloor$ is the largest integer that is not greater than x.

From Theorem 1, when M is relatively large, $\mathcal{L}_u \subseteq \mathcal{M}(\overline{\nu}_u)$ and K is greater than and close to $2MW$, the detection result of (17) is asymptotically equal to that of the MMSE detector.

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Theorem 2: When $K = \lfloor 2MW(1-\epsilon) \rfloor$, $0 < \epsilon < 1$ and $\mathcal{M}(\overline{\nu}_u) \cap \mathcal{L}_{u'} = \emptyset$, we have

$$\lim_{M\to\infty} \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}} \mathbf{h}_{u'} = \mathbf{0}_K. \tag{21}$$

From Theorem 2, when M is relatively large and K is slightly less than $\lfloor 2MW \rfloor$, if $\mathcal{M}(\overline{\nu}_u)$ and $\mathcal{L}_{u'}$ do not overlap, $\mathbf{d}_1(\overline{\nu}_u), \cdots, \mathbf{d}_K(\overline{\nu}_u)$ are asymptotically orthogonal to $\mathbf{h}_{u'}$, i.e., $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u) \approx \mathbf{0}_K$. Under these conditions, $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ can be omitted in (16) for complexity reduction.

With the aid of Theorem 1 and Theorem 2, the parameter settings are given as follows.

1) W and $\overline{\nu}_u$: According to Theorem 1, we set

$$W \geq \overline{W}, \quad \overline{\nu}_u = \frac{1}{2}\left(\nu_u^{\mathrm{l}} + \nu_u^{\mathrm{r}}\right), \tag{22}$$

where $\overline{W} \triangleq \frac{1}{2}\max_{u\in\mathcal{U}}\{\nu_u^{\mathrm{r}} - \nu_u^{\mathrm{l}}\}$ is half of the maximum length of $\mathcal{L}_1, \cdots, \mathcal{L}_U$. It can be checked that $\mathcal{L}_u \subseteq \mathcal{M}(\overline{\nu}_u)$ holds for each UT u.

2) K: Under the condition $\mathcal{L}_u \subseteq \mathcal{M}(\overline{\nu}_u)$, we discuss the value selection of K. It can be known from Theorem 1 and Theorem 2 that the detection performance and the detector simplification cannot be guaranteed at the same time. Nevertheless, it can be known from the simulation that when K is slightly less than $\lfloor 2MW \rfloor$, the detection performance loss is insignificant. For this reason, we compromise a small fraction of detection accuracy for complexity reduction by setting K less than and close to $2MW$. Hence, we set a threshold λ^{th} which is less than and close to 1, and K can be selected as the maximum integer that satisfies $\lambda_K > \lambda^{\mathrm{th}}$, i.e.,

$$K = \max_{K'\in\mathcal{A}, \lambda_{K'} > \lambda^{\mathrm{th}}} K'. \tag{23}$$

After setting parameters by (22) and (23), (16) can be simplified as

$$\mathbf{r}_u^{\mathrm{sst}} \approx \left(\sum_{u'\in\overline{\mathcal{U}}_u} \overline{\mathbf{h}}_{u'}(\overline{\nu}_u)\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)^{\mathrm{H}} + \frac{1}{\rho}\mathbf{I}_K \right)^{-1} \overline{\mathbf{h}}_u(\overline{\nu}_u), \tag{24}$$

where

$$\overline{\mathcal{U}}_u \triangleq \left\{ u' \middle| \mathcal{M}(\overline{\nu}_u) \cap \mathcal{L}_{u'} \neq \emptyset,\ u' \in \mathcal{U} \right\} \text{ with } \overline{U}_u = |\overline{\mathcal{U}}_u|. \tag{25}$$

Additionally, by exploiting the sparse beam domain channel, $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ can be expressed as

$$\overline{\mathbf{h}}_{u'}(\overline{\nu}_u) = \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{V}\tilde{\mathbf{h}}_{u'} = \mathbf{G}(\overline{\nu}_u)\tilde{\mathbf{h}}_{u'} = \mathbf{G}_{u'}(\overline{\nu}_u)\hat{\mathbf{h}}_{u'}, \tag{26}$$

where $\hat{\mathbf{h}}_u \triangleq \mathbf{N}_u\tilde{\mathbf{h}}_u \in \mathbb{C}^{B_u}$ represents the reduced-dimensional beam domain channel with non-zero elements of $\tilde{\mathbf{h}}_u$, $\mathbf{G}(\overline{\nu}_u) \triangleq \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{V} \in \mathbb{C}^{K\times\tilde{M}}$, $\mathbf{G}_{u'}(\overline{\nu}_u) \triangleq \mathbf{G}(\overline{\nu}_u)\mathbf{N}_{u'}^{\mathrm{T}} \in \mathbb{C}^{K\times B_{u'}}$ and $\mathbf{N}_u \in \mathbb{R}^{B_u\times\tilde{M}}$ is the beam selection matrix with

$$[\mathbf{N}_u]_{i,j} \triangleq \begin{cases} 1, & j = [\mathcal{B}_u]_i \\ 0, & \text{else.} \end{cases} \tag{27}$$

In (26), the computation of $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ requires the knowledge of $\mathbf{G}(\overline{\nu}_u)$. According to (8), $\mathcal{L}_u$ changes only when statistical CSI $\tilde{\boldsymbol{\omega}}_u$ changes. It should also be noted that $\tilde{\boldsymbol{\omega}}_u$ varies much more slowly than instantaneous CSI $\tilde{\mathbf{h}}_u$, thus $\mathbf{D}(\overline{\nu}_u)$ and $\mathbf{G}(\overline{\nu}_u)$ can be pre-calculated for each channel realization.

We refer to the detector discussed above as the separate Slepian transform (SST) based detector since the Slepian transform of $\mathbf{y}$ is done for each UT independently. Algorithm 1 concludes the design procedure for this detector.

Algorithm 1: SST based detector

Input : $\hat{\mathbf{h}}_u, \rho, \mathbf{D}(\overline{\nu}_u), \mathbf{G}(\overline{\nu}_u), \mathbf{N}_u, \overline{\mathcal{U}}_u$
Output: $\mathbf{R}^{\mathrm{sst}}$
1 Compute $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ $(\forall u' \in \overline{\mathcal{U}}_u, u \in \mathcal{U})$ by (26).
2 Compute $\mathbf{r}_1^{\mathrm{sst}}, \cdots, \mathbf{r}_U^{\mathrm{sst}}$ by (24).
3 Compute $\mathbf{R}^{\mathrm{sst}}$ by (18).

D. Joint Slepian Transform Based Detector

In the previous subsection, the Slepian transform is performed for each UT separately and modulated Slepian sequences are designed accordingly, which is relatively cumbersome and challenging to implement when the UT number is large. For this reason, a fixed set of modulated Slepian sequences are considered in this subsection.

Construct $\mathbf{D}(\tilde{\nu}_1), \cdots, \mathbf{D}(\tilde{\nu}_L)$ (M is divisible by L) which are independent of specific channel realizations. For the sake of simplicity, we set $\tilde{\nu}_1, \cdots, \tilde{\nu}_L$ at equal intervals on $[-1/2, 1/2)$ and symmetry about the origin, i.e.,

$$\tilde{\nu}_l = \frac{1}{L}\left(l - \frac{1}{2}\right) - \frac{1}{2}. \tag{28}$$

To guarantee the detection performance, according to Theorem 1, the following condition should be satisfied,

$$\forall u \in \mathcal{U}, \exists l_u \in \{1, \cdots, L\},\ \text{s.t. } \mathcal{L}_u \subseteq \mathcal{M}(\tilde{\nu}_{l_u}). \tag{29}$$

To this end, we set

$$W \geq \overline{W} + \frac{1}{2L},\ l_u = \left\lfloor L\left(\overline{\nu}_u + \frac{1}{2}\right)\right\rfloor + 1, \tag{30}$$

so that (29) is met. Under these conditions, K is set by (23) for the same reason.

After determining $\mathbf{D}(\tilde{\nu}_l)$, by applying the Slepian transform to $\mathbf{y}$, the l-th compressed observation vector is given by

$$\begin{aligned} \overline{\mathbf{y}}(\tilde{\nu}_l) &= \sum_{u\in\tilde{\mathcal{U}}_l} \overline{\mathbf{h}}_u(\tilde{\nu}_l)x_u + \sum_{u\in\mathcal{U}\setminus\tilde{\mathcal{U}}_l} \overline{\mathbf{h}}_u(\tilde{\nu}_l)x_u + \overline{\mathbf{z}}(\tilde{\nu}_l) \\ &\overset{(a)}{\approx} \sum_{u\in\tilde{\mathcal{U}}_l} \overline{\mathbf{h}}_u(\tilde{\nu}_l)x_u + \overline{\mathbf{z}}(\tilde{\nu}_l), \end{aligned} \tag{31}$$

where

$$\tilde{\mathcal{U}}_l \triangleq \left\{ u \middle| \mathcal{M}(\tilde{\nu}_l) \cap \mathcal{L}_u \neq \emptyset,\ u \in \mathcal{U} \right\} \text{ with } \tilde{U}_l = |\tilde{\mathcal{U}}_l|, \tag{32}$$

and (a) follows from Theorem 2.

With the compressed observation vector in (31), the detector of UT u is given by

$$\mathbf{r}_u^{\mathrm{jst}} = \arg\min_{\mathbf{r}_u\in\mathbb{C}^K} \mathbb{E}\left\{ \left| x_u - \mathbf{r}_u^{\mathrm{H}}\overline{\mathbf{y}}(\tilde{\nu}_{l_u}) \right|^2 \right\} = \mathbf{C}_{l_u}^{-1}\overline{\mathbf{h}}_u(\tilde{\nu}_{l_u}) \in \mathbb{C}^K, \tag{33}$$

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where

$$\mathbf{C}_l = \sum_{u\in\tilde{\mathcal{U}}_l} \overline{\mathbf{h}}_u(\tilde{\nu}_l)\overline{\mathbf{h}}_u(\tilde{\nu}_l)^{\mathrm{H}} + \frac{1}{\rho}\mathbf{I}_K \in \mathbb{C}^{K\times K}. \quad (34)$$

The computation of $\overline{\mathbf{h}}_u(\tilde{\nu}_l)$ involved in (33) and (34) can also be expressed as

$$\overline{\mathbf{h}}_u(\tilde{\nu}_l) = \mathbf{G}_u(\tilde{\nu}_l)\hat{\mathbf{h}}_u = \mathbf{G}(\tilde{\nu}_l)\mathbf{N}_u^{\mathrm{T}}\hat{\mathbf{h}}_u, \quad (35)$$

where $\mathbf{G}(\tilde{\nu}_l) = \mathbf{D}(\tilde{\nu}_l)^{\mathrm{H}}\mathbf{V}$ is fixed and independent of channel statistics, hence can be computed offline.

We call the detector the joint Slepian transform (JST) based detector since the Slepian transform (31) is applied for UTs in $\mathcal{N}_l = \{u | l_u = l,\ u \in \mathcal{U}\}$ collectively. The design process of the JST based detector is summarized in Algorithm 2.

Algorithm 2: JST based detector

Input : $\hat{\mathbf{h}}_u, \rho, \mathbf{D}(\tilde{\nu}_l), \mathbf{G}(\tilde{\nu}_l), \mathbf{N}_u, \tilde{\mathcal{U}}_l$
Output: $\mathbf{r}_1^{\mathrm{jst}}, \cdots, \mathbf{r}_U^{\mathrm{jst}}$
1 Compute $\overline{\mathbf{h}}_u(\tilde{\nu}_l)$ $(u \in \tilde{\mathcal{U}}_l, l = 1\cdots, L)$ by (35).
2 Compute $\mathbf{C}_1, \cdots, \mathbf{C}_L$ by (34).
3 Compute $\mathbf{r}_1^{\mathrm{jst}}, \cdots, \mathbf{r}_U^{\mathrm{jst}}$ by (33).

Armed with designed detector in Algorithm 2, the detection result is presented as

$$\hat{\mathbf{x}}^{\mathrm{jst}} = \left[\left(\mathbf{r}_1^{\mathrm{jst}}\right)^{\mathrm{H}}\overline{\mathbf{y}}(\tilde{\nu}_{l_1}), \cdots, \left(\mathbf{r}_U^{\mathrm{jst}}\right)^{\mathrm{H}}\overline{\mathbf{y}}(\tilde{\nu}_{l_U})\right]^{\mathrm{T}}. \quad (36)$$

Note that $\overline{\mathbf{y}}(\tilde{\nu}_l)$ in (36) needs to be computed in the detection process. By utilizing the structural properties of $\mathbf{D}(\tilde{\nu}_l)$, the complexity of this computational process can be further reduced. Construct

$$\begin{aligned}\mathbf{Q} &\triangleq [\mathbf{D}(\tilde{\nu}_1), \cdots, \mathbf{D}(\tilde{\nu}_L)] \\ &= [\mathrm{diag}\{\mathbf{d}_1(\tilde{\nu}_1)\}, \cdots, \mathrm{diag}\{\mathbf{d}_K(\tilde{\nu}_1)\}]\,(\mathbf{I}_K \otimes \boldsymbol{\Omega})\mathbf{P}^{\mathrm{T}}, \quad (37)\end{aligned}$$

where $\otimes$ is the Kronecker product operator. $\mathbf{P} \in \mathbb{R}^{KL\times KL}$ is the permutation matrix with $\mathbf{e}_{KL, K\langle i-1\rangle_L + \lfloor (i-1)/L\rfloor + 1}$ in its i-th column, where $\mathbf{e}_{N,n}$ denotes the n-th column of $\mathbf{I}_N$, $\langle\cdot\rangle_N$ indicates the modulo-N operation, and

$$\boldsymbol{\Omega} \triangleq [\boldsymbol{\omega}(0), \boldsymbol{\omega}(1/L), \cdots, \boldsymbol{\omega}((L-1)/L)] = \mathbf{1}_{M/L} \otimes \mathbf{F}_L^{\mathrm{H}}, \quad (38)$$

where $\mathbf{1}_N$ denotes N-dimensional column vector with all 1s, and $\mathbf{F}_L$ represents L-dimensional DFT matrix, i.e., $[\mathbf{F}_L]_{i,j} = e^{-\bar{\jmath}2\pi(i-1)(j-1)/L}$. It can be shown that

$$\begin{aligned}\overline{\mathbf{y}}(\tilde{\nu}_l) &= \left(\mathbf{e}_{L,l}^{\mathrm{T}} \otimes \mathbf{I}_K\right)\mathrm{vec}\left\{[\overline{\mathbf{y}}(\tilde{\nu}_1), \cdots, \overline{\mathbf{y}}(\tilde{\nu}_L)]\right\} \\ &= \left(\mathbf{e}_{L,l}^{\mathrm{T}} \otimes \mathbf{I}_K\right)\mathbf{Q}^{\mathrm{H}}\mathbf{y} \\ &= \left(\mathbf{e}_{L,l}^{\mathrm{T}} \otimes \mathbf{I}_K\right)\mathbf{P}\mathrm{vec}\left\{\mathbf{F}_L\left[\boldsymbol{\zeta}_1, \cdots, \boldsymbol{\zeta}_K\right]\right\}, \quad (39)\end{aligned}$$

where $\mathrm{vec}\{\cdot\}$ denotes the vectorizing operation, and

$$\begin{aligned}\boldsymbol{\zeta}_i &\triangleq \left(\mathbf{1}_{M/L}^{\mathrm{T}} \otimes \mathbf{I}_L\right)\left(\mathbf{d}_i(\tilde{\nu}_1)^* \odot \mathbf{y}\right) \\ &= \sum_{r=1}^{M/L}\left(\mathbf{e}_{M/L,r}^{\mathrm{T}} \otimes \mathbf{I}_L\right)\left(\mathbf{d}_i(\tilde{\nu}_1)^* \odot \mathbf{y}\right). \quad (40)\end{aligned}$$

Therefore, the computation of $\overline{\mathbf{y}}(\tilde{\nu}_l)$ can be fast implemented by L-dimensional FFT. The JST based detection process is summarized in Algorithm 3.

Algorithm 3: JST based detection

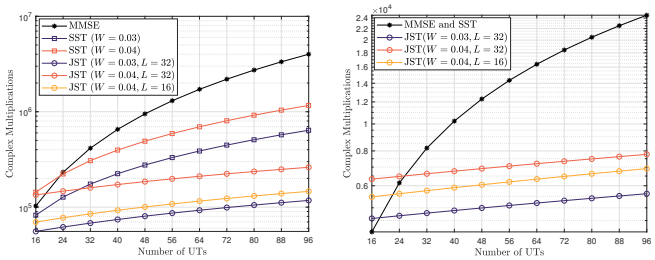
Input : $\mathbf{y}, \mathbf{r}_u^{\mathrm{jst}}, \mathbf{D}(\tilde{\nu}_1)$
Output: $\hat{\mathbf{x}}^{\mathrm{jst}}$
1 Compute $\boldsymbol{\zeta}_1, \cdots, \boldsymbol{\zeta}_K$ by (40).
2 Compute $\overline{\mathbf{y}}(\tilde{\nu}_1), \cdots, \overline{\mathbf{y}}(\tilde{\nu}_L)$ by (39).
3 Compute $\hat{\mathbf{x}}^{\mathrm{jst}}$ by (36).

E. Complexity analysis

We discuss the computational complexity of the proposed detectors in terms of required complex multiplications. To this end, define the average value of B_u, $\overline{U}_u$ and $\tilde{U}_l$ as $B \triangleq \frac{1}{U}\sum_{u\in\mathcal{U}} B_u$, $\overline{U} \triangleq \frac{1}{U}\sum_{u\in\mathcal{U}}\overline{U}_u$, $\tilde{U} \triangleq \frac{1}{L}\sum_{l=1}^{L}\tilde{U}_l$, respectively. Table I summarizes the theoretical bounds of design complexity and detection complexity of different detectors, respectively. Fig. 1 plots the results under the given system configuration with different parameter settings. It can be observed that for the proposed detectors, the advantage of design complexity becomes rather apparent as the number of UTs increases, and the advantage of detection complexity of the JST based detector is obvious.

TABLE I
COMPUTATIONAL COMPLEXITY OF EACH DETECTOR

Detector	Design complexity	Detection complexity
MMSE	$\frac{(U+1)}{2}+\frac{U^3}{2}+\frac{3U^2}{2}+MU^2$	MU
SST	$\left(\left(B+\frac{(\overline{K}+1)}{2}\right)\overline{U}+\frac{\overline{K}^2}{2}+\frac{5\overline{K}}{2}+M\right)KU$	MU
JST	$\left(\left(B+\frac{(\tilde{K}+1)}{2}\right)\tilde{U}+\frac{\tilde{K}^2}{2}+\frac{3\tilde{K}}{2}\tilde{K}\right)KL+K^2U$	$K(M+U+\frac{L}{2}\log_2 L)$



(a) Design complexity (b) Detection complexity

Fig. 1. Comparison of (a) design complexity (b) detection complexity against UT number (from 16 to 96). Parameters are set as $M = 256$, $\lambda^{\mathrm{th}} = 0.99$ $(K = \lfloor 2MW\rfloor - 2)$.

IV. Simulation Results

In this section, we provide simulation results to show the performance of detectors for HF skywave massive MIMO-OFDM systems.

A. Simulation Setup

In the simulation, we consider HF skywave massive MIMO-OFDM systems and the parameters are given in Table II.

We utilize Proplab-Pro version 3.1, a commercial ray-tracing software, to generate realistic HF skywave channel

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TABLE II
SIMULATION PARAMETERS

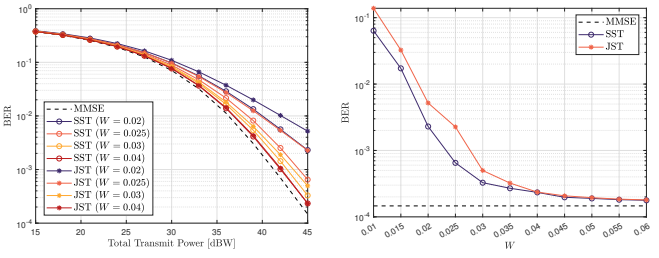
Parameter	Value	Parameter	Value
Carrier frequency f_c	16 MHz	Antenna number at BS M	256
Subcarriers spacing Δf	250 Hz	Antenna spacing at BS d	9 m
Subcarrier number N_c	2048	Valid subcarrier number N_v	1536
Modulation scheme	16QAM	UT number U	96

parameters. The corresponding beam domain channel and statistical CSI are acquired by the method proposed in [11] with $\tilde{M} = 2M$.

B. Detection Performance

To evaluate the performance of proposed detectors, we discuss the bit error rate (BER) versus the total transmit power ϵ (in dBW) without channel coding.

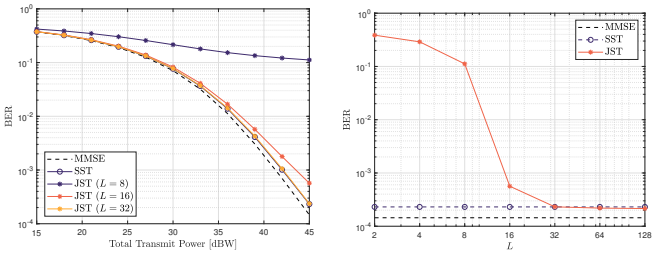
In Fig. 2, the detection performance of proposed detectors with different values of W are compared. From the figure, the detection performances for both the SST based detector and the JST based detector improve as W increases, and are closely approaching that of the MMSE detector. Moreover, when W is large, the detection performance of the JST based detector is almost the same as that of the SST based detector, while when W is small, the detection performance of the JST based detector has a more significant gap than that of the SST based detector. It can be inferred that when W is small, the condition $\mathcal{L}_u \subseteq \mathcal{M}(\tilde{\nu}_{l_u})$ can not be well satisfied, resulting in a performance gap between the JST based detector and the SST based detector. However, the complexities of proposed detectors increase as W increases since the effect of dimensionality reduction is not remarkable when W is large. Therefore, W should be as large as possible with acceptable complexity.



(a) (b)

Fig. 2. BER performances comparison against (a) the total transmit power ϵ (b) W (ϵ = 45 dBW), and (λ^{th}, L) = (0.99, 32).

Fig. 3 compares the detection performance of the JST based detector with different values of L. The results of SST based detector and the MMSE detector are also included as baselines. From the complexity analysis, we know that smaller L results in lower detection complexity, while the detection performance deteriorates when L decreases, as shown in the figure. Specifically, $L \leq 8$ yields extremely unacceptable detection performance as (30) is not met. Consequently, the trade-off between detection performance and complexity can be balanced with $L = 32$.



(a) (b)

Fig. 3. BER performances comparison against (a) the total transmit power ϵ (b) L (ϵ = 45 dBW), and (W, λ^{th}) = (0.04, 0.99).

V. CONCLUSION

In this paper, we have investigated the detector based on Slepian transform for HF skywave massive MIMO-OFDM. We have introduced a beam based massive MIMO-OFDM channel model for HF skywave systems. By exploiting beam domain sparsity of the channel model, we designed modulated Slepian sequences for each UT and performed Slepian transform on the observation vector and the channel matrix for dimensionality reduction, and the SST based detector is proposed. Next, we developed the JST based detector with fixed modulated Slepian sequences for further complexity reduction. Simulation results demonstrate that by choosing appropriate parameters, the proposed detectors can provide similar performance to that of the MMSE detector at a much lower implementation cost.

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